

PLOUMANAC'H PERROS GUIREC, France

WMO: 071170

Lat: 48.8259N

Lon: 3.4735W

Elev: 190

StdP: 14.60

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	33.7	36.0	24.7	18.9	36.5	27.4	21.5	38.5	52.3	51.1	47.4	50.3	11.5	120	0.619

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.1	72.9	64.5	70.4	63.5	68.8	62.7	66.1	70.5	64.8	68.8	63.7	67.5	9.5	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.3	90.8	67.9	63.1	87.0	66.8	62.1	83.9	65.9	30.9	70.6	29.9	69.0	29.1	67.6	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
36.5	29.3	24.9	DB	29.3	82.3	3.2	5.2	27.0	86.0	25.1	89.1	23.4	92.0	21.0	95.8	
			WB	27.4	69.2	3.6	2.0	24.8	70.7	22.8	71.9	20.8	73.0	18.2	74.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.1	46.2	45.6	48.0	50.9	55.0	59.4	62.8	63.2	61.4	57.3	51.4	47.4
	DBStd	7.42	4.75	4.62	4.30	3.67	3.38	3.09	2.82	2.57	3.14	4.06	3.95	4.86
	HDD50	519	131	132	89	29	1	0	0	0	0	3	28	106
	HDD65	4018	583	542	528	423	312	171	80	68	116	242	407	546
	CDD50	2015	13	10	26	56	155	282	396	411	342	228	71	25
	CDD65	39	0	0	0	0	1	3	11	13	8	3	0	0
	CDH74	78	0	0	0	0	4	9	21	24	15	5	0	0
	CDH80	15	0	0	0	0	1	1	5	4	3	1	0	0
(14) Wind	WSAvg	12.1	14.6	14.6	13.4	11.5	10.9	10.0	9.5	10.0	10.9	11.7	13.7	14.4
(15) Precipitation	PrecAvg	33.9	3.7	3.3	2.2	2.6	2.3	2.0	2.0	2.2	1.8	3.5	4.1	4.3
	PrecMax	44.0	6.7	6.4	6.4	6.8	4.5	4.6	4.6	5.8	5.3	7.5	8.0	9.3
	PrecMin	24.2	0.8	0.6	0.3	0.8	0.6	0.3	0.8	0.7	0.2	0.6	1.1	0.9
	PrecStd	4.5	1.7	1.6	1.3	1.5	1.0	1.1	1.0	1.1	1.2	1.6	1.8	1.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.1	59.2	63.5	66.7	69.9	74.4	76.2	76.7	75.2	72.2	63.0	59.4
		MCWB	54.0	53.3	54.3	57.1	61.0	65.8	65.3	66.3	64.2	63.0	58.8	54.6
	2%	DB	55.6	55.7	58.2	61.5	65.5	69.3	71.8	71.9	70.9	67.2	60.5	57.3
		MCWB	53.3	51.6	52.6	54.4	58.8	62.7	64.3	64.9	63.5	61.2	56.9	54.5
	5%	DB	54.2	53.7	55.7	58.5	62.5	66.8	69.7	69.6	68.3	64.7	58.3	55.6
		MCWB	52.0	50.6	51.8	52.8	57.3	61.2	63.3	63.4	62.7	60.0	55.2	53.3
	10%	DB	52.7	52.1	53.8	56.5	60.3	64.7	68.1	68.0	66.4	62.7	56.8	54.0
		MCWB	50.5	49.5	50.6	52.0	55.7	59.9	62.4	62.5	61.4	59.1	54.0	51.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.1	54.6	55.7	57.7	62.7	66.5	67.2	68.4	67.0	64.3	60.2	56.5
		MCDB	55.9	57.2	59.6	64.0	68.4	72.9	72.8	74.0	71.3	68.8	61.9	57.5
	2%	WB	53.9	52.6	54.0	55.3	59.8	63.6	65.5	66.1	65.0	62.3	57.4	55.0
		MCDB	55.3	54.5	56.8	60.1	64.1	68.1	70.2	69.8	69.0	65.8	59.4	56.5
	5%	WB	52.4	51.2	52.4	53.8	57.8	61.8	64.3	64.7	63.7	60.9	55.8	53.6
		MCDB	53.8	53.2	55.0	57.3	61.5	65.7	68.5	68.0	67.2	63.5	57.7	55.3
	10%	WB	51.0	49.8	50.9	52.5	56.5	60.5	63.2	63.5	62.4	59.6	54.5	52.1
		MCDB	52.4	51.8	53.4	55.8	59.7	63.9	67.0	66.7	65.7	62.2	56.5	54.1
(35) Mean Daily Temperature Range	5% DB	MDBR	6.9	7.5	8.6	9.7	9.9	9.6	10.1	10.1	9.5	8.7	7.1	6.8
		MCDBR	7.6	9.7	11.5	12.9	13.0	12.7	12.3	12.5	12.3	11.1	8.7	7.5
	5% WB	MCWBR	7.4	7.2	7.5	7.8	7.8	7.6	6.8	7.0	7.2	6.9	7.1	7.2
		MCWBR	7.2	8.2	10.4	11.5	11.9	11.7	11.5	11.2	11.0	9.7	8.3	7.5
(40) Clear-Sky Solar Irradiance	taub	taub	0.333	0.348	0.376	0.387	0.391	0.384	0.383	0.373	0.363	0.357	0.335	0.326
		taud	2.409	2.391	2.324	2.317	2.354	2.399	2.405	2.453	2.463	2.466	2.442	2.373
	Ebn at Noon	Ebn at Noon	228	253	264	273	276	278	277	275	266	247	226	212
		Edh at Noon	21	27	34	37	37	36	36	32	29	25	20	19
(44) All-Sky Solar Radiation	RadAvg	RadAvg	296	541	897	1403	1674	1835	1757	1484	1161	668	364	247
	RadStd	RadStd	25	63	93	130	132	195	103	118	84	52	36	30

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page